



Business Case Study: Leak Detection

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Table of Contents

01

Problem-Definition

Overview

Limitations and Assumptions

02

Proposed Solutions

Technology-People-Process

03

Implementation

Immediate Action

Short-Term

Long-Term

04

Anticipated Results

Repeatability and
Reproducibility

Metrics of Success



01 Problem Definition

Problem Overview

Condenser units are experiencing refrigerant leaks leading to customer dissatisfaction and loss of revenue.



5% of units
produced found
to have leaks



Operator Fatigue
from Sensitive
Equipment



\$2.5 million per
year loss in
warranty returns



Products are not
meeting EPA
regulations

Possible Causes

Technology

- High-sensitivity sniffer wand
- Improper leak detection booth

People

- Operator fatigue from continuous alarm
- Insufficient training with new equipment (reset button)

Process

- Low-cycle time for leak detection
- Inadequate circulation/venting step
- Buildup of tracer gas

Limitations and Assumptions

- **How is the sniffer wand calibrated?**

Assumed reset sets the threshold to a new minimum baseline

- **How is air circulated/vented from the booth?**

Assumed a fan system circulates air within the booth

- **How well are leaks detected by the supplier?**

Assumed this was not possible to measure

- **Are there issues in the brazing process?**

Assumed this was outside of the scope of this study

- **What is cost of downtime and brazing repair?**

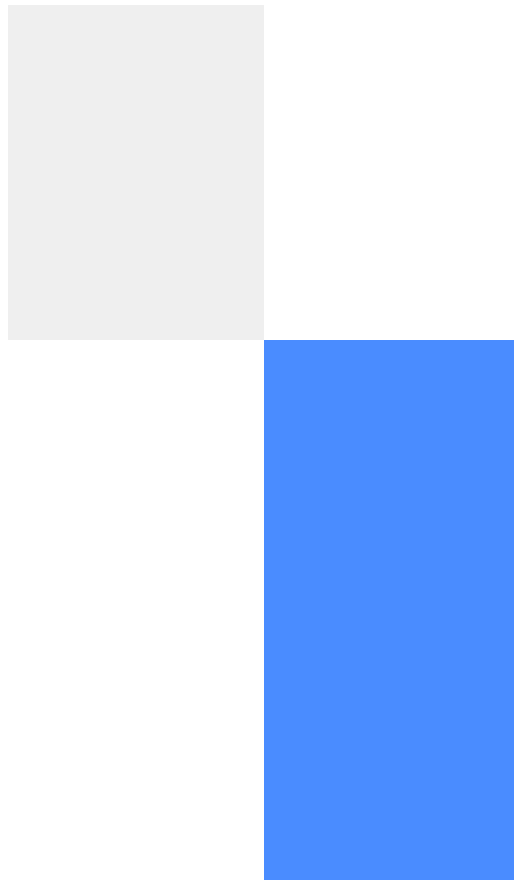
Assumed this was too high to justify a higher cycle time

- **Are leaks being created during the shipping process?**

Assumed this was a next step

02

Proposed Solutions



Technology

Sniffer Wand

- **Too expensive to replace** immediately
- Needs to be properly **calibrated** to reduce false positives
- Could be improved through **automated machinery, ultrasonic detection, and pressure-drop tests** in long-term future to reduce human error

Detection Booth

- Needs to be properly purged with a **fume hood or vent** and air circulation per the leak detection study
- Fume hood installation could cost around **\$6,000 to \$60,000 per unit** depending on the size of the booth (National Laboratory Sales)

People

Training

- Operators need to be **retrained to use the new sniffer** wand equipment as reset button could cause false negatives
 - Can be done immediately at little to no cost
- Should understand how to **use air circulation and venting** to reduce contamination
 - Circulation system ideally would not involve excess operator training

Process

- Continuous vs. Periodic **cycling of air** circulation and venting
 - Decrease contaminants while reducing cycle time and leak bleed without human intervention
- A new study could be performed to assess **where leaks are most common**
 - operators could check problem areas first and more closely
 - Would only slightly increase cycle time but increase efficiency
- Conduct a **reliability and reproducibility** study
 - determine if the optimized scanning is efficient and consistent across operators

03 Implementation Process

Long-Term

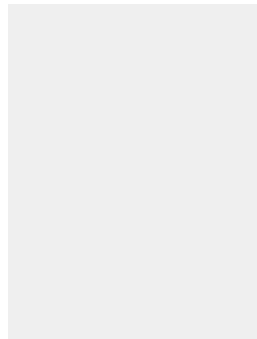
Short-Term

Immediate Action

- Retrain Operators
- Improve air circulation per study
- Install fume hoods or vents
- Optimized scanning path
- Train operators on venting
- Implement automated or ultrasonic leak detection
- Find other root causes in manufacturing or brazing processes
- Conduct study into other trace gases

04

Anticipated Results



Metrics



- A **repeated study** of leak detection to determine if the decontamination and retraining is effective
 - Conducted under similar time pressure and with a variety of operators
 - Measure percent of undetected leaks below $1.0\text{E-}6 \text{ cm}^3/\text{s}$
 - Increase in detections that pass EPA regulations
- A Gage R&R study to determine **consistency and reliability** with optimized scanning and gas purging
 - Keep cycle time at or below 90s
 - Reduce number of faulty units shipped to less than 5%
 - Decrease in revenue lost through warranty below \$2.5 million
- Eliminate irritation over **alarms** from sensitive equipment
 - Improved operator morale and function

Thank You

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